DAVID ADEOGUN

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PROFESSIONAL SUMMARY

Artificial Intelligence / Machine Learning Engineer with 4+ years in ML and AI, including 7+ years of software engineering, specializing in model development, training, debugging, optimization, and evaluation across applied research and production systems. Designed and deployed end-to-end ML pipelines spanning computer vision, natural language processing, time-series analysis, and robotics perception, from synthetic data generation to production deployment and monitoring. Experienced building scalable ML infrastructure on AWS (SageMaker, ECS, S3) and Kubernetes, with a demonstrated ability to bridge cutting-edge research and production-grade engineering.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, LSTMs, CNNs, Vision Transformers, AdaBoost, YOLO, XGBoost, LightGBM, Random Forests, Neural Networks, ONNX Runtime, TensorFlow Serving, Hugging Face Transformers, Stable Diffusion, Reinforcement Learning, Azure OpenAI, Prompt Engineering, Structured Output Parsing

Experiment & Evaluation: MLflow, Weights & Biases, Experiment Tracking, Model Registry, Hyperparameter Tuning, Optuna, Ray Tune, A/B Evaluation, Benchmark Design, RMSE/Precision/Recall/F1 Metrics, Cross-Validation, Feature Importance Analysis, SHAP

ML Infrastructure & Ops: AWS (SageMaker, ECS, ECR, S3, Lambda), Kubernetes, Docker, Apache Airflow, Kubeflow, CI/CD for ML Pipelines, GitHub Actions, Automated Rollbacks, Model Versioning, DVC, Feature Stores, Production Monitoring, Prometheus, Grafana, Google Cloud (Cloud Run, Cloud SQL, Firebase Hosting), Spring AI

Programming & Data: Python, C++, SQL, MATLAB, NumPy, Pandas, Apache Kafka, MQTT, Spark, PostgreSQL, Redis, REST APIs, FastAPI

Perception & Robotics: OpenCV, PCL, ROS/ROS2, LiDAR Processing, 3D Point Cloud Segmentation, Sensor Fusion, SLAM, Computer Vision, Depth Estimation, Object Detection

INDUSTRY EXPERIENCE

Artificial Intelligence/ Machine Learning Engineer | Mode Robots Oct 2023 – Present

- Designed and trained production machine learning models across supervised, unsupervised, and semi-supervised paradigms including LSTMs, CNNs, Vision Transformers, XGBoost, and AdaBoost, achieving measurable improvements in prediction accuracy, false alarm reduction, and model generalization across structured and unstructured datasets.
- Engineered scalable feature extraction and selection pipelines across high-dimensional structured datasets, applying dimensionality reduction, normalization, and encoding techniques to improve model accuracy and reduce computational overhead across multiple production ML systems.
- Developed and maintained synthetic data generation pipelines using NumPy and Pandas, producing 150,000+ labeled training samples with controlled noise injection and distribution shifts to improve model robustness and generalization in low-data environments.
- Fine-tuned and evaluated large language models for domain-specific reasoning and classification tasks, designing schema-driven evaluation pipelines using AWS Bedrock to benchmark response quality, groundedness, and decision-logic across frontier models.
- Applied systematic hyperparameter optimization using Optuna and Ray Tune, improving key model performance metrics including RMSE, precision, recall, and F1 scores, impacting model reliability and business decision accuracy across production systems.
- Designed and evaluated agentic AI systems using prompt engineering, tool-use frameworks, and chain-of-thought reasoning techniques, benchmarking model behavior across multi-step reasoning tasks and integrating guardrails, output validation, and fallback logic to ensure reliability and safety in production AI deployments.
- Architected and deployed AI-driven decision-support systems combining large language models, structured reasoning pipelines, and retrieval-augmented generation (RAG) to deliver explainable, context-aware recommendations for complex business scenarios, reducing manual decision-making time and improving consistency across high-stakes operational workflows.
- Implemented full experiment tracking, model versioning, and model registry workflows using MLflow and Weights & Biases, ensuring complete reproducibility across training runs, enabling rapid rollback and audit trails for all production models.
- Containerized and deployed scalable ML inference endpoints using Docker, AWS ECS, and Kubernetes, optimizing TensorFlow Serving and ONNX Runtime configurations to meet production latency and throughput requirements at scale.
- Built end-to-end ML pipelines spanning data ingestion, feature engineering, model training, evaluation, and deployment, reducing model development cycle time and ensuring consistency across research and production environments using PyTorch, scikit-learn, and AWS SageMaker.

AI Application Engineer | Natterbase Feb 2022 – Sep 2023

- Designed and built an AI-powered cognitive-behavioral decision support system (ED² Insurance Choice) integrating Azure OpenAI GPT-4o for personalized risk assessment across 7 insurance domains, enabling users to evaluate coverage alternatives through structured decision-making and problem-solving modes.
- Engineered structured prompt templates with dynamic placeholder injection, constructing domain-specific prompts that combine user profile data, insurance coverage alternatives, and actuarial guidelines to generate personalized risk-related outcomes with confidence levels.
- Implemented hallucination mitigation strategies including temperature-zero inference, source constraint instructions, and explicit citation requirements in prompts, reducing fabricated outputs and ensuring GPT-generated risk values (Baseline Risk, Relative Risk Reduction, Residual Risk) are grounded in verifiable data sources.
- Developed a response caching layer in PostgreSQL that stores GPT outputs with user profile snapshots, eliminating outcome inconsistency across repeated interactions by returning cached results when profile data is unchanged and triggering fresh GPT calls only on profile changes.
- Built a full-stack production application with Angular 13 frontend, Spring Boot backend, and PostgreSQL database, implementing JWT authentication, CORS configuration, and RESTful API design to support secure, stateful user sessions.
- Designed and implemented a preprogrammed 7-field User Profile Template based on insurance industry risk assessment factors, serving as the single source of truth for GPT personalization and Baseline Risk calculation across all insurance domains.
- Created a domain template mapping system storing problem definitions, future risk events, and 3-tier coverage alternatives for 7 insurance types, enabling automatic screen population and prompt construction without manual user input.
- Integrated Azure OpenAI GPT-4o via Spring AI, parsing structured JSON responses into domain-specific UI fields including risk reduction metrics, difficulty categories with solutions, and significance categories with strategies.
- Deployed the application to Google Cloud (Cloud Run for backend, Firebase Hosting for frontend, Cloud SQL for PostgreSQL), configuring environment-based secrets management for API keys and database credentials across development and production environments.
- Resolved production deployment issues including Spring Security CORS configuration conflicts, JWT authentication endpoint permissions, and database seeding for subscription plans, ensuring seamless operation across desktop and mobile platforms.
- Addressed 6 core AI reliability research problems - hallucination, session history loss, outcome inconsistency, unclear data retention requirements, opaque AI sources, and source inconsistency through a combination of prompt engineering, caching architecture, and API configuration strategies.

Software Engineer | Williams Sonoma Inc Jun 2020 – Jan 2022

- Engineered and scaled backend services for catalog, cart, and checkout workflows, supporting high-traffic production systems and ensuring system reliability during peak loads.
- Developed and maintained vendor-facing web applications enabling real-time tracking of customer order fulfillment across warehouse locations, improving supply chain visibility and reducing order status inquiry resolution time.
- Built and optimized backend services supporting order lifecycle management between distribution centers, ensuring data accuracy and reliability for high-volume retail operations across vendor and warehouse touchpoints.
- Designed and implemented RESTful API integrations between internal inventory management systems and third-party logistics providers, enabling automated stock synchronization and reducing manual data reconciliation by 40%.
- Collaborated with cross-functional teams to identify and resolve production performance bottlenecks, optimizing database queries and caching strategies that improved API response times by 30% during seasonal traffic surges.

Software Developer | Finance Life Technologies Aug 2015 - May 2020

- Developed and maintained core modules for an enterprise lending platform supporting customer onboarding, loan approval workflows, and loan servicing, enabling financial access for underserved and unbanked populations.
- Built and integrated backend services for automated loan processing pipelines, improving approval turnaround time and ensuring regulatory compliance across customer lifecycle stages from application to repayment.
- Designed and implemented role-based access control and audit logging across the lending platform, ensuring data security and regulatory traceability for sensitive financial transactions and customer records.
- Developed automated reporting modules that aggregated loan portfolio metrics including disbursement volumes, repayment rates, and delinquency trends, enabling management to make data-driven decisions on lending strategies.
- Built and maintained integration services connecting the lending platform with external credit bureaus and payment gateways, automating credit scoring retrieval and real-time payment processing for loan disbursements and collections.

RESEARCH EXPERIENCE

Kennesaw State University – *Advanced Mechatronics and Intelligent Robotics Lab*

Contributed to an NIH-funded research project focused on the design and fabrication of soft parallel robots for transcatheter cardiovascular interventions. The research addresses a critical need in cardiac surgery, heart disease kills one person every 34 seconds in the United States, and transcatheter procedures offer a significantly less invasive alternative to open-heart surgery.The project develops miniature soft robotic devices capable of navigating the cardiovascular system to perform precise surgical interventions, combining novel soft actuator design, physics-based modeling, and master-slave control architectures to give surgeons enhanced dexterity and haptic feedback during catheter-based procedures.

- Robotics Researcher: Developing next-generation soft parallel robots for transcatheter cardiovascular interventions under a National Institutes of Health (NIH) grant.
- Modelling & Control: Developed physics-based models using MATLAB, Simulink, and Simscape to simulate complex 6-DOF robotic movements.
- System Architecture: Designed robust control architectures for master-slave robotic systems to enhance surgical precision and haptic feedback.

Digital Twin–Driven Predictive Maintenance for Industrial Robotics Using IIoT Telemetry and Machine Learning

Developed an end-to-end predictive maintenance framework integrating digital twin simulation, synthetic IIoT data generation, and comparative machine learning evaluation to detect equipment degradation and predict remaining useful life (RUL) in robotic manufacturing cells.

- Designed and validated a digital twin of an ABB IRB 1200 pick-and-place robotic cell in Siemens Tecnomatix Process Simulate, establishing the IIoT-to-ML data pipeline architecture for Industry 4.0 predictive maintenance research.
- Engineered a synthetic telemetry generation pipeline producing 150,000+ labeled operational cycles across five sensor channels (motor current, temperature, vibration, vacuum pressure, cycle time) with calibrated Gaussian noise and three physics-based degradation modes.
- Trained and benchmarked LSTM time-series networks against AdaBoost baseline classifiers, achieving 5,000+ cycle early fault detection lead time, <5% false alarm rate, and RUL prediction RMSE under 2,000 cycles.
- Demonstrated that sequential temporal modeling (LSTM on 500-cycle sliding windows) captures gradual wear patterns invisible to snapshot-based classifiers, quantifying the accuracy-latency tradeoffs for time-series predictive maintenance applications.

PREPRINTS

[1] Adeogun, D., Vu, K., Alam, W., Quansah, E. “Digital Twin–Driven Predictive Maintenance for Pick-and-Place Robotic Cells Using IIoT Telemetry and Machine Learning.” ASME IMECE 2026. (To Be Submitted) https://doi.org/10.13140/RG.2.2.15447.53926

[2] Adeogun, D., Sandoval, S., Wulfekuhle, V., Vu, K., Kouwenhoven, F. “Improving Efficiency of Deep-Learning based Vision Transformer for Remote Sensing for Satellite Images.” ResearchGate, January 2026. https://doi.org/10.13140/RG.2.2.13764.49282

EDUCATION

M.S. Intelligent Robotic Systems | Kennesaw State University

- *Relevant Coursework: Machine Learning, Machine Vision, Artificial Intelligence, Perception/Navigation/Path Planning, Robot Simulation, 3D Point Cloud Manipulation*

B.S. Software Development | Brigham Young University-Idaho